

Homework #1 for Geospatial Analysis & Remote Sensing (EDS223 - Fall 2026)

2026-10-10

```
# Load in Packages
library(tidyverse)
library(sf)
library(here)
library(tmap)

# Read in geodatabase of EJScreen data at the Census Block Group level
ej <- sf::st_read(here::here(
  "data", "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"))

# Filter dataset to a state
louisiana <- ej |>
  dplyr::filter(ST_ABBREV == "LA")

# Filter dataset to a county
east_baton <- louisiana |>
  dplyr::filter(CNTY_NAME == "East Baton Rouge Parish")

# Find the average values for all variables within counties
ks_counties <- aggregate(louisiana, by = list(louisiana$CNTY_NAME),
  FUN = mean)

# Cleaning data so we do not have NA columns in the interactive map
clean_ks <- ks_counties |>
  select(-c("ID", "STATE_NAME", "ST_ABBREV", "CNTY_NAME"))
```

```

# Make the map output static
tmap_mode("plot")

# Map for low income community percentage and houses build before 1960

tm_shape(clean_ks) +
  tm_polygons(fill = "LOWINCPCT",
    fill.scale = tm_scale(values = "-viridis"),
    fill.legend = tm_legend(title = "Low Income Community (proportion)",
      position = tm_pos_out("right", "center"))) +

  tm_symbols(fill = "PRE1960PCT",
    size = 0.3,
    shape = 21,
    col = "black",
    lwd = 0.5,
    fill.scale = tm_scale(values = c("lightpink", "darkmagenta")),
    fill.legend = tm_legend(
      title = "Houses Built Before 1960\n(proportion)",
      position = tm_pos_out("right", "center"))) +

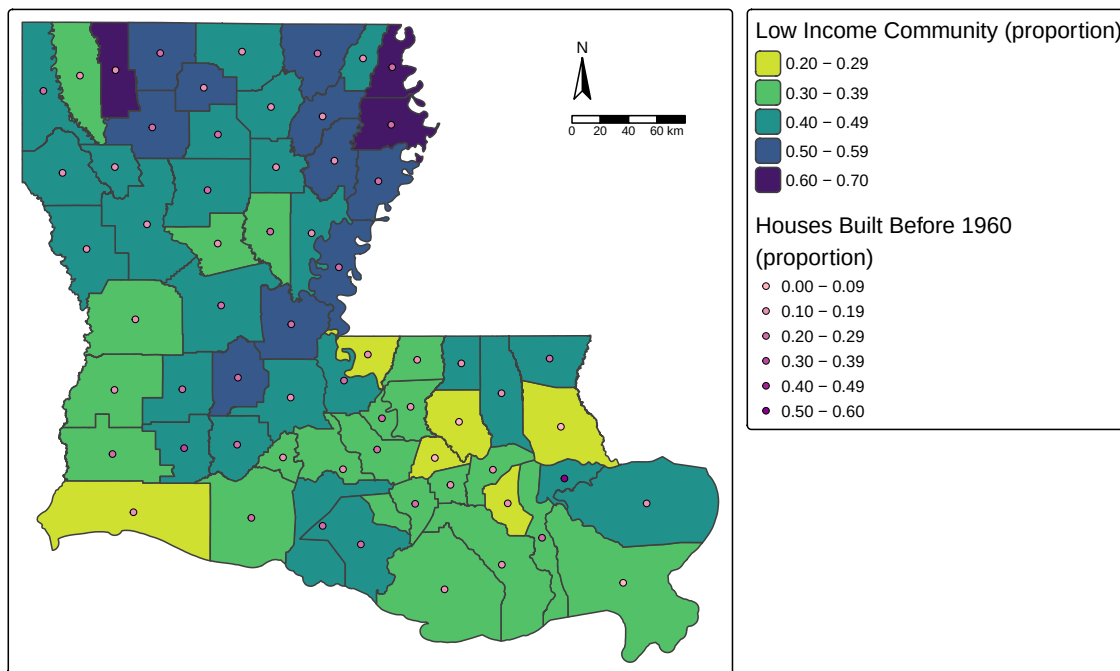
  tm_compass(size = 2, text.size = 0.7, position = c("right", "top")) +

  tm_scalebar(width = 10, text.size = 0.5, position = c("right", "top")) +

  tm_title("Low Income Communities in Houses\nBuilt Before 1960 in Louisiana",
    size = 1.0,
    position = tm_pos_out("center", "top", pos.h = "left", pos.v = "bottom"))

```

Low Income Communities in Houses Built Before 1960 in Louisiana



```
# Map for low income community percentage and particulate matter

tm_shape(east_baton) +
  tm_polygons(fill = "PEOPCOLORPCT",
    fill.scale = tm_scale(values = "brewer.blues"),
    fill.legend = tm_legend(title = "People of Color (proportion)",
      position = tm_pos_out("right", "center")))) +

  tm_symbols(fill = "PM25",
    size = 0.3,
    shape = 21,
    col = "black",
    lwd = 0.5,
```

```

    fill.scale = tm_scale(values = c("orange", "darkred")),
    fill.legend = tm_legend(title = "PM2.5 ( $\mu\text{g}/\text{m}^3$ )",
    position = tm_pos_out("right", "center")) +

tm_compass(size = 2, text.size = 0.7, position = c("left", "bottom")) +

tm_scalebar(width = 10, text.size = 0.5, position = c("left", "bottom")) +

tm_title(
  "People of Color Exposed to Particulate Matter\n
  in East Baton Rouge Parish, LA",
  size = 1.0,
  position = tm_pos_out("center", "top", pos.h = "left", pos.v = "bottom"))

```

People of Color Exposed to Particulate Matter

in East Baton Rouge Parish, LA

